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Advanced Mud Gas Logging Techniques for Hydrogen Exploration

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Abstract

The exploration of natural hydrogen or "white hydrogen" has gained significant attention due to its potential as a clean and sustainable energy source. Unlike industrially produced hydrogen, which relies on energy-intensive processes such as steam methane reforming or electrolysis, natural hydrogen is generated through geochemical and biological processes within the Earth's crust. It can be continuously produced through reactions such as serpentinization, radiolysis of water, and deep subsurface and soil biological activity.

This study investigates various degassing and gas analysis techniques for hydrogen detection while drilling, including real-time mass spectrometry at the wellsite and gas chromatography with thermal conductivity detection (GC-TCD). Particular attention is given to the calibration of mud-logging gas chromatographs equipped with mass spectrometry (GC-MS) to enable accurate quantification of hydrogen at low concentrations (e.g., below 5 ppm). Additionally, advanced analytical technologies such as cavity-enhanced Raman spectroscopy are explored as potential improvements for enhancing sensitivity and selectivity in hydrogen detection.

A major challenge in hydrogen logging is the artificial generation of hydrogen when drilling with oil-based mud, which can be mitigated by analyzing alkene and carbon monoxide concentrations obtained from continuous logging or spot sampling. Addressing these challenges, the study proposes a workflow for quantitative hydrogen logging, contributing to the assessment of this resource in hybrid drilling scenarios, including oil and gas, geothermal applications, and dedicated hydrogen exploration wells.

Introduction

The exploration of natural hydrogen, commonly referred to as white or geological hydrogen, has attracted increasing attention as a potential clean and renewable energy resource. In contrast to industrial hydrogen, which is typically produced through energy-intensive methods such as steam methane reforming or electrolysis, natural hydrogen is continuously generated in the Earth's crust through a combination of geochemical and microbial processes. This inherent generation capability positions it as a low-carbon alternative with promising economic viability (Zgonnik, 2020).

The most prominent geochemical mechanisms responsible for hydrogen production include serpentinization, wherein ultramafic rocks react with water to release hydrogen via iron oxidation, and

radiolysis, in which ionizing radiation from naturally occurring radioactive elements dissociates water molecules within crystalline basement rocks. Subsurface microbial activity may also contribute to hydrogen accumulation, either via fermentation or via methanogenesis, under specific geochemical conditions.

Since the early 2020s, interest in natural hydrogen has expanded globally, with reported exploration efforts in Mali, Russia, Djibouti, Australia, France, Colombia, and the United States. The hypothesis of continuous in-situ generation raises the prospect of naturally replenishing hydrogen reservoirs, underscoring the need for reliable detection and quantification methodologies during drilling operations.

Despite its potential, several technical challenges hinder accurate assessment of natural hydrogen during drilling. Its low solubility in drilling fluids may lead to underestimation of concentrations, while artificial hydrogen generation, particularly in wells drilled with oil-based muds under high-temperature conditions and harsh drilling regimes, can result in false positives. Drill-bit metamorphism (DBM) and chemical reactions involving drilling additives further complicate interpretation, necessitating the development of adapted analytical strategies.

This study presents a comprehensive evaluation of hydrogen detection techniques employed during drilling, integrating field observations and laboratory simulations. It focuses on gas degassing behavior, calibration of gas chromatographs equipped with mass spectrometers (GC-MS), and controlled DBM experiments. A practical workflow is proposed for quantitative hydrogen logging across conventional hydrocarbon wells, geothermal applications, and dedicated hydrogen exploration, supporting the advancement of reliable and scalable subsurface hydrogen assessment.

Natural Hydrogen Generation Processes

As introduced earlier, natural hydrogen in the subsurface is primarily produced through two mechanisms: serpentinization, the reaction of ultramafic rocks with water under reducing conditions, and radiolysis, the dissociation of water molecules by ionizing radiation from naturally occurring uranium (U), thorium (Th), and potassium (K). These processes operate under distinct thermal and geological conditions and may generate hydrogen continuously over geologic timescales (Figure 1).

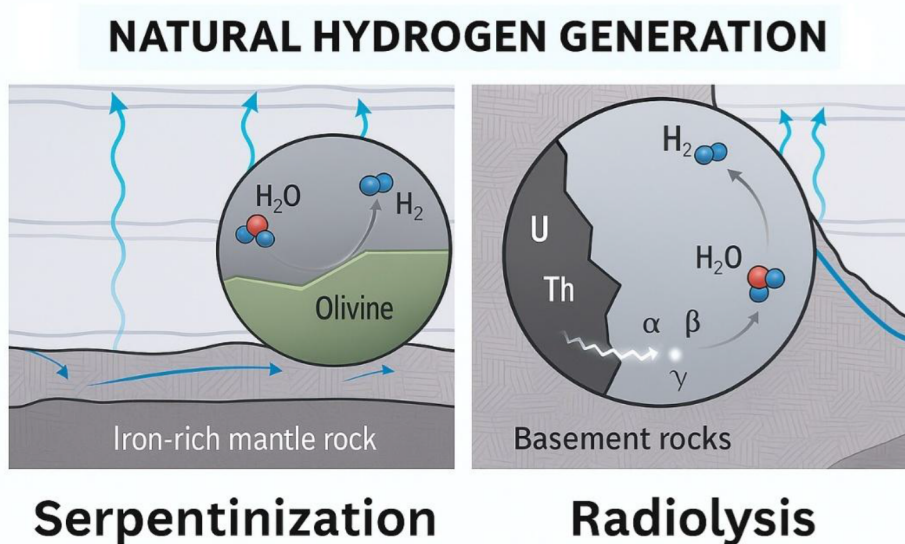


Figure 1—Conceptual illustration of subsurface hydrogen generation mechanisms. Water-rock interaction with olivine releases H_2 (Left); radioactive decay in basement rocks splits H_2O into H_2 and O_2 (Right).

Serpentinization typically occurs in ophiolitic complexes, forearc zones, and ultramafic intrusions, where mantle-derived rocks are exposed to circulating fluids. Radiolytic hydrogen, by contrast, is associated with

Precambrian basement rocks, especially those enriched in radioactive elements and characterized by low permeability. Additional favorable settings include fractured crystalline terrains and geothermal systems, where structural and thermal conditions promote both hydrogen generation and migration.

Although early-stage exploration often relies on surface seep detection, soil gas surveys, and geophysical methods, these techniques provide limited insight into deep subsurface hydrogen systems. Mud logging, traditionally developed for hydrocarbon detection, offers a continuous and depth-resolved record of formation gases during drilling. Recent advancements in gas extraction, sensor technologies, and data analytics have transformed standard mud logging into a more advanced and flexible platform capable of detecting trace gases, including hydrogen (Strapoc et al. 2022), at ppm levels with improved temporal and spatial resolution.

Integrating hydrogen monitoring into advanced mud logging workflows provides a high-impact opportunity to identify active hydrogen generation zones, particularly in wells not initially intended for hydrogen exploration. When combined with mineralogical and geochemical analysis of cuttings, such as iron oxidation states, serpentinization minerals, or U-Th-K concentrations, this approach enables more robust interpretation of subsurface processes and supports the identification of previously unrecognized hydrogen plays.

The following section examines the operational and analytical challenges that constrain hydrogen detection using gas extracted during drilling and proposes strategies to overcome these limitations in both conventional and hydrogen-focused drilling environments.

Natural Hydrogen Logging Challenges Using Mud Logging Techniques

Applying conventional mud logging techniques to detect natural hydrogen presents a distinct set of technical and operational challenges. These challenges arise from hydrogen's physical properties, such as low molecular weight, high diffusivity, and limited solubility, as well as from the complexity of geological settings in which it is encountered. While mud gas extraction during drilling is a well-established practice in hydrocarbon exploration, its extension to hydrogen detection requires careful consideration of several fundamental limitations (Zgonnik, 2020).

Low Solubility in Drilling Fluids

Hydrogen exhibits solubility in drilling fluids that is approximately equivalent to methane and becomes higher in reservoir water below about 1.5 km (Lopez-Lazaro et al. 2019). Under surface conditions, hydrogen dissolves in water-based and oil-based muds at concentrations typically around 1 ppm by weight. This limited solubility promotes the enhanced release of hydrogen into the gas phase upon return to surface, thereby facilitating its extraction by conventional degassing systems. Although it enhances gas-phase transfer at the surface, the low solubility of hydrogen also implies an increased risk that a significant fraction may be lost before reaching detection systems, particularly in open circulation environments or under turbulent flow conditions.

High-Frequency Variability and Limited Depth Resolution

Hydrogen migration in the subsurface is often controlled by localized structural features such as fractures and fault conduits. These small-scale pathways can produce sharp and transient concentration peaks. However, conventional gas logging systems typically lack the temporal resolution and sensitivity needed to detect such narrow anomalies, especially during fast-rate of penetration drilling, leading to missed or mischaracterized hydrogen-rich intervals (Prinzhofer et al., 2019; Zgonnik, 2020).

Instability in Surface Circulation Systems

The surface handling environment also contributes to uncertainty in hydrogen detection. Due to its small molecular size and rapid diffusivity, hydrogen readily escapes through leaks, interfaces, or open mud

systems. Variability in fluid circulation rates, degasser efficiency, and sampling design further complicates gas retention and quantification. Without careful control of these variables, hydrogen measurements may become erratic or systematically underestimated (Sherwood Lollar et al., 2014).

Analytical Limitations in Gas Detection Equipment

Most wellsite gas analyzers currently in use are configured for hydrocarbon detection and are not optimized for hydrogen quantification. These instruments often lack hydrogen-specific calibration, and their detection ranges may not capture the full spectrum of expected hydrogen concentrations, which can range from below 1 ppm to several volume percent, depending on the formation and depth. This mismatch leads to poor sensitivity, particularly at low concentrations (<10 ppm), and undermines the ability to confidently distinguish hydrogen anomalies from background levels (Larin et al., 2015). To address these challenges, suitable analytical technologies with low detection limits and high stability are essential for accurate hydrogen detection during drilling.

Analytical Technologies for Hydrogen Detection in Mud Gas Logging

Detecting natural hydrogen during drilling presents unique challenges due to its low solubility, rapid diffusivity, and susceptibility to artificial generation through high-temperature drilling processes. Conventional mud logging tools, optimized for hydrocarbon systems, often lack the sensitivity, specificity, or resolution needed to detect and differentiate hydrogen at low concentrations. These issues highlight the need for advanced detection workflows based on high-resolution analytical systems capable of capturing trace and transient hydrogen signals across complex operational environments.

Accurate detection and quantification of hydrogen (H_2) in mud gas streams require dedicated analytical systems capable of handling low detection limits, broad dynamic ranges, and complex gas mixtures. Reliable quantification also requires consistent and matrix-relevant calibration to account for sensor response and variability across concentrations. This section provides a detailed evaluation of the main technologies currently in use or emerging for wellsite hydrogen analysis, with emphasis on their suitability for real-time mud logging applications.

Direct Quadrupole Mass Spectrometry (DQ-MS)

Direct quadrupole mass spectrometry (DQ-MS) offers compact, carrier-gas-free detection of multiple gases based on their mass-to-charge (m/z) ratios. Its real-time measurement capability and ability to simultaneously monitor permanent gases, such as hydrogen, helium, methane, and carbon dioxide, make it an attractive option for remote wellsite operations.

However, hydrogen detection with DQ-MS faces intrinsic analytical challenges. The key limitation lies in the mass resolution required to accurately distinguish low-mass species. Hydrogen ($m/z = 2$) and helium ($m/z = 4$) are separated by narrow mass intervals and may be affected by background interferences or adjacent peaks (e.g., H^+ fragments, water cluster ions), particularly when the analyzer lacks high-resolution capability. These interferences can introduce uncertainty, especially at trace concentrations.

Moreover, the diversity of gas mixtures encountered during drilling, from ambient background gases to complex mud gas emissions, necessitates multivariate calibration strategies. Single-point calibration is generally insufficient due to matrix effects and the potential overlap of signals. Robust performance requires comprehensive calibration across a range of hydrogen concentrations and compositions, including interference-prone mixtures, to achieve accurate quantification and minimize misidentification.

While DQ-MS remains a valuable tool for screening applications and operational diagnostics, its use in quantitative hydrogen logging demands careful attention to instrument resolution, matrix calibration, and potential mass interferences.

Gas Chromatography–Mass Spectrometry (GC-MS)

Gas Chromatography–Mass Spectrometry (GC-MS) integrates chromatographic separation with molecular-level mass detection, offering a highly selective and adaptable platform for wellsite gas analysis. It is uniquely positioned to meet the dual requirements of conventional hydrocarbon logging and emerging hydrogen exploration, owing to its ability to quantify hydrogen, helium, carbon dioxide, and hydrocarbons (C1–C5) in a single analytical cycle.

A major advantage of GC-MS lies in its operational flexibility. Unlike traditional chromatographic systems that depend on helium or argon as carrier gases, GC-MS can be configured to operate with nitrogen or even filtered ambient air. This capability greatly reduces logistical complexity during field operations, particularly in remote or offshore environments where the availability of high-purity specialty gases is limited.

To confirm the suitability of GC-MS for hydrogen logging, we conducted a dedicated calibration campaign using certified hydrogen gas mixtures across a wide range of concentrations (5 ppm to 10 vol %). As shown in Figure 2, the calibration procedure validated both the sensitivity and dynamic range of the system. Mass-selective chromatographic peaks for hydrogen were perfectly resolved at 5 ppm and 100 ppm, demonstrating unambiguous separation from neighboring permanent gases such as helium and carbon dioxide. The full-range response curve (Figure 2, left) exhibited excellent linearity across several orders of magnitude, confirming the robustness of the method and the reproducibility of the instrument response.

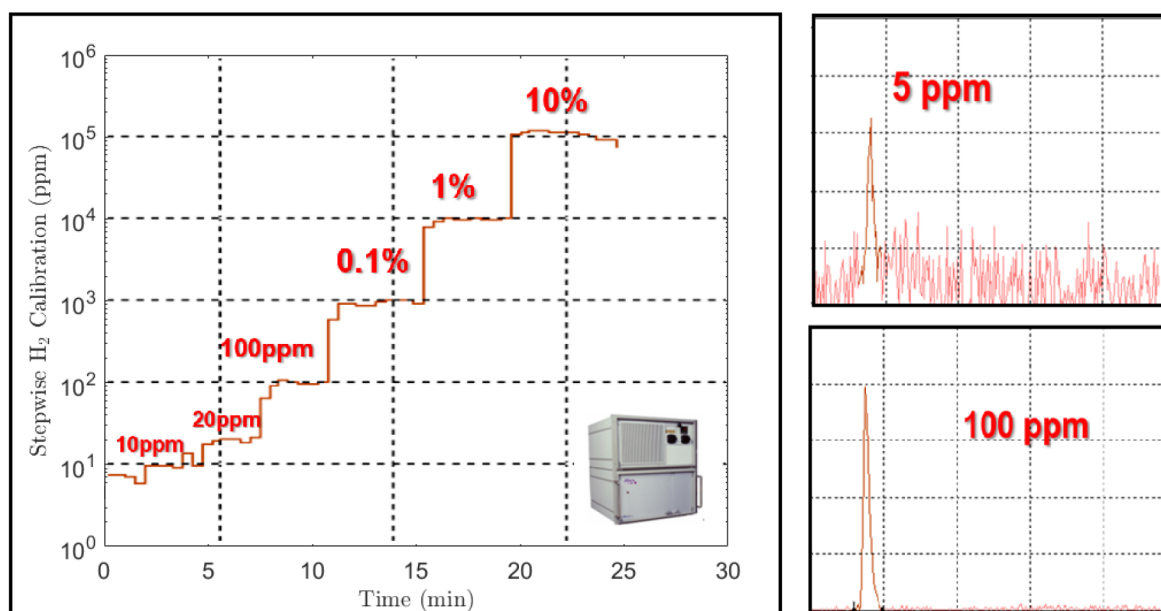


Figure 2—Stepwise calibration of hydrogen using GC-MS across a wide concentration range (5 ppm to 10 vol%). The left panel shows the dynamic response curve, illustrating linearity over five orders of magnitude. The chromatograms on the right confirm mass-selective identification of hydrogen at 5 ppm and 100 ppm, with clear separation from other permanent gases. The image of the GC-MS system used for this calibration is shown within the main panel.

The image inset (Figure 2, top-right) shows the field-deployed GC-MS system used in these experiments, highlighting the practicality of this analytical workflow under typical wellsite conditions. This wellsite-ready configuration featured inert sample pathways and low dead volume to preserve hydrogen integrity throughout the analytical cycle. The achieved limit of detection, consistently below 5 ppm, is sufficient to track subtle geologic hydrogen anomalies and capture transient changes in concentration during dynamic drilling operations.

Importantly, GC-MS also provides additional diagnostic value beyond hydrogen detection. Its ability to detect helium is particularly valuable in hydrogen exploration, as helium is often co-generated via radiolytic

processes and can serve as a geochemical tracer of deep-seated hydrogen generation mechanisms. Likewise, the simultaneous detection of carbon dioxide supports broader interpretations of fluid-rock interactions. However, the quantification of carbon monoxide remains challenging due to its interaction with mud, and when using nitrogen- or air-based carrier gas systems (i.e., N_2 and CO have the same nominal masses).

When compared to Direct Quadrupole Mass Spectrometry (DQ-MS), GC-MS offers several analytical advantages. Both methods rely on mass-selective detection; however, DQ-MS lacks the chromatographic separation step, leading to potential overlap in the low-mass region where hydrogen, water, nitrogen, and carbon monoxide fragments can interfere. DQ-MS also requires matrix-specific or multivariate calibration strategies to compensate for ion suppression effects and compositional variability in mixed gas streams. In contrast, GC-MS maintains compositional integrity through physical separation before detection, reducing matrix effects and enabling more straightforward calibration.

Although GC-MS and DQ-MS exhibit similar analytical cycle times (typically 60–90 seconds), the former imposes superior specificity, robustness, and quantitative reliability. These attributes make it a strong candidate for routine hydrogen detection during drilling operations, especially when combined with standard hydrocarbon analysis in integrated workflows.

Gas Chromatography–Thermal Conductivity Detection (GC-TCD)

Gas chromatography with thermal conductivity detection (GC-TCD) is widely used for analyzing permanent gases such as hydrogen and helium. The method operates by measuring the difference in thermal conductivity between the sample gas and the carrier gas—typically high-purity argon or helium—as analytes elute from the column. Because hydrogen and helium have significantly higher thermal conductivity than most other gases, they generate distinct and quantifiable signals.

GC-TCD is well suited for field deployment due to its mechanical robustness, low maintenance requirements, and rapid analysis cycles (typically 20–60 seconds). Its linear response across a wide concentration range supports routine hydrogen monitoring, particularly in dedicated exploration settings where fast, standalone screening is advantageous.

In complex gas mixtures, column extensions can be used to improve analyte separation. Carrier gas selection may be adapted to the application: helium enhances sensitivity for hydrocarbons, while argon allows simultaneous detection of hydrogen and helium. In some configurations, dual-column setups are employed to broaden analytical coverage. These attributes make GC-TCD a reliable and efficient tool for real-time hydrogen detection in wellsite operations.

Complementary and Emerging Technologies

Chemoelectrical Sensors. Chemoelectrical hydrogen sensors detect hydrogen based on changes in electrical conductivity upon exposure. These sensors are valued for their rapid response times, often less than one second, and their ease of installation directly on degasser outlets or gas extraction points. Their compact design and low power requirements make them suitable for continuous monitoring during drilling operations.

However, chemoelectrical sensors typically operate within a limited dynamic range and are optimized for detecting low hydrogen concentrations. Their performance can be affected by cross-sensitivity to other reducing gases (e.g., methane, CO), and they often require frequent recalibration to maintain reliability. As such, they are most effectively used as supplementary tools or for triggering alarms rather than as primary analytical systems.

Cavity-Enhanced Raman Spectroscopy (CERS). CERS is an emerging optical technique that enhances Raman scattering signals using a resonant optical cavity. By exciting gases with a high-power laser and increasing the intracavity photon density, this method allows for the detection of trace gases such as

hydrogen, carbon monoxide, and carbon dioxide at concentrations as low as 10–100 ppm. Integration times typically range from 10 to 100 seconds, depending on the target gas and environmental conditions.

The main advantages of CERS include its high selectivity, lack of need for carrier gases, and insensitivity to water vapor interference, making it particularly attractive for real-time applications in wet or complex drilling environments. However, current limitations in system durability, alignment sensitivity, and portability have so far restricted its widespread deployment at the wellsite. Continued development may address these challenges, potentially enabling broader use of CERS in hydrogen exploration.

Comparative Assessment. The main performance characteristics of analytical technologies used for hydrogen detection in mud logging are summarized in Table 1. GC-MS offers the most comprehensive solution, combining broad gas coverage (H₂, He, C1–C5), high quantification accuracy, and operational flexibility when using nitrogen or ambient air as a carrier gas. GC-TCD provides faster analysis cycles and reliable performance for permanent gases, though it depends on high-purity argon and is less suited for hydrocarbon detection.

Table 1—Summary of analytical technologies for hydrogen detection in mud logging, comparing detection capabilities, operational requirements, quantification accuracy, and field suitability

Analyzer Type	Detects H ₂	Detects He	Detects C1-C5	Carrier Gas Required	Analysis Cycle	Concentration Range	Quantification Accuracy	Wellsite Suitability	Key Limitations
GC-MS	✔ Yes	✔ Yes	✔ Yes	✔ No (Air or N ₂ optional)	60–90 s	~5 ppm to 10–20 vol%	✔ High	✔ Ideal	Moderate cycle time; requires calibration
GC-TCD + GC-FID	✔ Yes (TCD)	✔ Yes (TCD)	✔ Yes (FID)	✘ Yes (Argon + H ₂)	20–60 s	Broad (ppm to vol%)	✔ High	✔ Ideal	Requires dual systems; sensitive to carrier gas quality
DQ-MS	✔ Yes	✔ Yes	✔ Yes	✔ No	60–90 s	Limited dynamic range	⚠ Moderate	✔ Compact	H ₂ signal amplification; limited low-range precision
Chemoelectrical Sensor	✔ Yes	✘ No	✘ No	✔ No	1–5 s	Low range only (ppm to ~0.5%)	⚠ Low	✔ Portable	No speciation; only detects H ₂ in narrow range
CERS (Raman-based)	✔ Yes	✔ Yes	✔ Yes (some)	✔ No	10–100 s	10 ppm to % range	✔ High	⚠ Experimental	Emerging technology; fragile optics; not yet field-proven for continuous logging

DQ-MS systems offer compact design and real-time analysis without the need for carrier gas, but their limited dynamic range and non-linear response reduce accuracy, especially at low hydrogen concentrations. Chemoelectrical sensors are valued for their portability and fast response times, making them effective for safety monitoring; however, they offer low accuracy due to interferences and no gas speciation. Cavity-Enhanced Raman Spectroscopy (CERS) delivers high selectivity without requiring a carrier gas, but its use in continuous field operations remains limited due to system fragility and alignment requirements.

Hydrogen Retention and Degassing Behavior

Hydrogen's behavior during mud logging challenges conventional assumptions based solely on equilibrium solubility. Under standard surface conditions, the solubility of hydrogen in water-based mud (WBM) is exceptionally low, on the order of 1 ppm by weight (Lopez-Lazaro et al., 2019). This value is nearly ten times lower than that of methane and far lower than carbon dioxide, which reaches ~1.5 g/L in water at 25 °C, but at reservoir depths becomes equivalent to methane. Based on these thermodynamic constraints, it has been widely assumed that hydrogen should escape immediately upon mud depressurization, thereby eluding detection by conventional surface degassers.

However, field observations have reported detectable hydrogen concentrations using standard mud logging setups, suggesting a more complex interplay between solubility, degassing kinetics, and surface

extraction efficiency. This section presents a combined field and laboratory investigation to better understand hydrogen retention and release mechanisms and assess the effectiveness of standard logging equipment.

Field Experimental Setup and Observations

A field experiment was designed to evaluate hydrogen extraction using two contrasting degassing configurations (Figure 3, left panel). The first configuration involved a gas trap mounted directly on the flowline ("Test"), intended to capture gases released immediately after the drilling mud exits the wellbore and undergoes a rapid pressure drop. The second configuration used a conventional agitator-based degasser ("Main") installed downstream of the shale shakers, representing standard practice in surface gas logging operations.

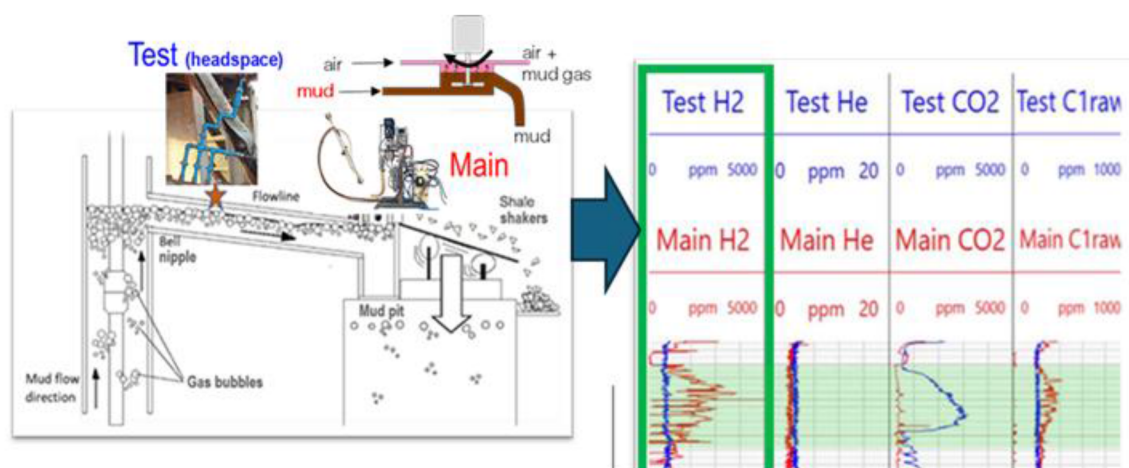


Figure 3—(Left) Field experiment comparing two hydrogen extraction configurations: a flowline-attached headspace trap (Test) and a downstream agitator-based degasser (Main). (Right) Gas log results show that hydrogen was detected only in Main, CO₂ was most intense in Test, and methane appeared in both. This illustrates the delayed release of hydrogen and the volatility of CO₂, adapted from Strapoc et al. 2022.

Gas logs obtained simultaneously from both extraction points (Figure 3, right panel) revealed a clear divergence in gas behavior. Hydrogen was consistently detected only in the Main configuration, with concentrations reaching several hundred ppm, suggesting it remained entrained in the mud and was released further downstream rather than immediately upon pressure drop. In contrast, carbon dioxide was dominant in the Test configuration, reflecting its high volatility and strong affinity for rapid phase transition at atmospheric pressure. Methane, which has intermediate solubility and diffusivity properties, was detected in both configurations, confirming its ability to transition to the gas phase under a wider range of surface conditions.

These results imply that hydrogen does not escape instantly at the surface, contrary to what would be expected from its low solubility alone. Instead, it appears to undergo delayed degassing, potentially due to kinetic constraints or physical retention mechanisms within the mud matrix. Conversely, carbon dioxide's early release may lead to partial loss if not captured near the flowline, suggesting that reliance on downstream degassing alone could result in significant underestimation, particularly in CO₂-rich formations.

Laboratory Validation and Physical Interpretation

To support and validate the field observations, a controlled laboratory experiment was conducted following drilling operations. After the well was completed, drilling mud samples containing hydrogen and methane were transferred to a surface facility, where they were circulated through a closed-loop degassing system designed to replicate surface handling conditions. The experiment monitored gas concentrations over time

as the mud was cycled repeatedly, simulating the degassing phase under ambient pressure and temperature, as shown in Figure 4 (left).

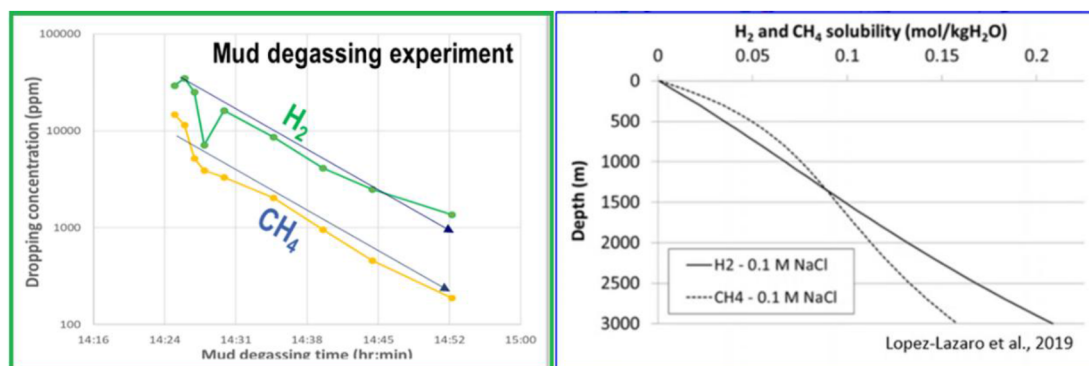


Figure 4—(Left) Laboratory degassing experiment showing parallel exponential decay of hydrogen and methane concentrations during surface circulation. (Right) Solubility-depth profiles for H_2 and CH_4 in saline mud, adapted from Lopez-Lazaro et al. (2019). CH_4 is more soluble than H_2 across depth, but the difference narrows with increasing pressure, following Henry's Law.

The results showed that both hydrogen and methane exhibited nearly identical exponential decline trends during circulation (degassing rate – Fig. 4, left), despite their substantial differences in equilibrium solubility. This behavior indicates that gas release is not governed solely by thermodynamic factors, but also by dynamic processes that hinder immediate degassing. These include kinetic limitations at the gas-liquid interface, which slow the nucleation and growth of gas bubbles, as well as physical retention mechanisms such as microbubble entrapment within the mud structure. In oil-based muds (OBM), surface tension and chemical interactions within the organic phase may further stabilize hydrogen, delaying its transition to the gas phase.

This interpretation is reinforced by solubility–depth profiles in saline brine (0.1 M NaCl), presented in Figure 4 (right). Although methane exhibits similar subsurface solubility in reservoir brines to hydrogen at all depths, below 1.5 km, the solubility of hydrogen becomes slightly higher. This pressure-dependent convergence is consistent with Henry's Law, which states that the solubility of a gas in a liquid increases proportionally with its partial pressure, and supports the view that both gases remain at least partially dissolved at depth. Upon returning to the surface, the release of hydrogen is limited not by its presence in the fluid but by the energy barrier required to initiate degassing, particularly in viscous or surfactant-rich mud systems.

Implications for Logging Workflows

These experimental results demonstrate that hydrogen can be reliably detected using conventional degassers, provided that retention dynamics are taken into account and appropriate calibration procedures are implemented. The results also highlight that CO_2 , due to its high volatility and solubility in water, requires dedicated headspace sampling near the flowline to avoid underestimation, an important consideration in geothermal or carbonate-rich environments. Methane exhibits intermediate behavior, more similar to H_2 , and proves useful as a reference gas for interpreting degassing kinetics and for validating the performance of gas extraction systems.

To enhance hydrogen detection in more demanding applications, such as high-resolution logging or reservoir-scale quantification, improvements to system design can be considered. These may include closed-loop degassing systems to maximize recovery efficiency, in-line probes or sensors positioned upstream of the main degasser to capture early-release gases, and refined extraction correction models that incorporate the distinct degassing kinetics of hydrogen compared to other gases.

Overall, this work reinforces the feasibility of incorporating hydrogen detection into existing mud logging operations and emphasizes the importance of integrating gas transport physics into the design and optimization of field instrumentation.

Artificial Hydrogen Generation During Drilling Operations

Distinguishing between hydrogen originating from geological formations and that produced artificially during drilling remains a critical challenge in natural hydrogen exploration. Several processes can generate hydrogen as an artifact of drilling operations, potentially leading to misinterpretation of mud gas logs. This section outlines key artificial hydrogen generation mechanisms, with emphasis on drill-bit metamorphism (DBM), chemical reactions involving hydrogen sulfide (H_2S) scavengers, and thermal degradation of organic drilling additives.

Drill-Bit Metamorphism (DBM)

Drill-Bit Metamorphism (DBM) refers to the localized thermal degradation of hydrocarbons at the bit–rock interface due to intense frictional heating during drilling. This effect is particularly pronounced when drilling with oil-based mud (OBM) in hard formations, at high rotary speeds, or with worn drill bits. Under these conditions, temperatures at the bit can exceed 250 °C and may reach over 1200 °C, triggering flash pyrolysis of long-chain aliphatic hydrocarbons (C_{12} – C_{20}) present in the OBM. This process results in the formation of gas-phase products, most notably hydrogen (H_2) and light alkenes such as ethene and propene, but also gaseous alkanes and carbon monoxide. These thermal signatures are consistent with laboratory flash pyrolysis results of hydrocarbons like pentadecane at 600 °C exposed to one second or less (Wenger et al., 2009; Herbinet et al., 2007).

Although hydrogen detected in the mud gas is often interpreted as an indicator of natural hydrogen migration from the subsurface, DBM introduces a non-formation-related source of hydrogen that can significantly distort gas measurements and their interpretation. Even with efficient extraction and accurate analytical techniques, the H_2 signal may be artificially elevated due to DBM, especially in OBM systems. Field data show that hydrogen concentrations in the range of several hundred to thousands of ppm can arise solely from OBM pyrolysis near the bit.

A defining characteristic of DBM is the strong correlation between hydrogen and light alkenes, as shown in Figure 5 (left). During thermal degradation, these gases are co-generated, and their concentrations tend to increase proportionally with DBM intensity. This relationship provides a valuable diagnostic basis: by monitoring ethene and propene, one can estimate the DBM-related hydrogen contribution.

The diagnostic use of alkene–hydrogen correlations and correction strategies were formalized by Strapáč et al. (2022), who demonstrated that DBM contributions can be quantified using field-collected gas data and removed from raw H_2 measurements to reveal the true formation signal. These DBM-correction workflows can be qualitatively applied in real-time at the rig site, quantitatively post-drilling via logging of alkenes or sealed spot samples such as IsoTubes, providing more reliable insight into subsurface hydrogen potential.

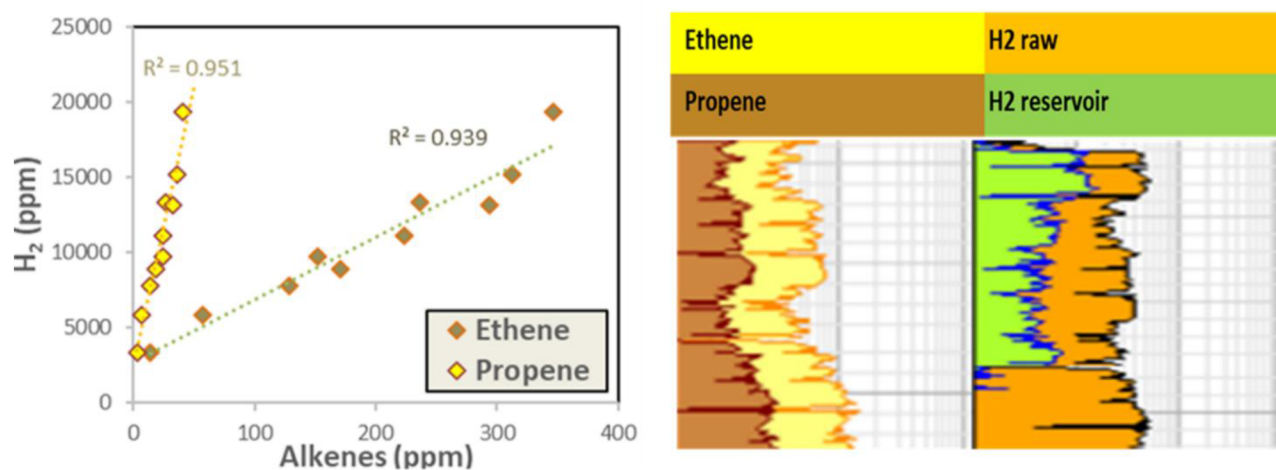


Figure 5—Strong correlation between hydrogen and alkenes (ethene, propene) generated by Drill-Bit Metamorphism (DBM) during OBM drilling of clean sandstone, adapted from [Strapoc et al. 2022](#). Left: H₂ vs. alkenes from mud gas spot samples from an experimental drilling. Right: Well log tracks showing raw and DBM-corrected H₂ using alkene-based correction.

Chemical Reactions Involving H₂S Scavengers

In sour gas drilling environments, hydrogen sulfide (H₂S) poses serious operational and safety challenges, necessitating the widespread use of chemical scavengers to neutralize its effects. Among these, triazine-based scavengers are particularly common due to their efficiency and compatibility with most drilling fluids. While these additives are designed to remove H₂S through stable chemical binding, recent studies have shown that under high-pressure, high-temperature (HPHT) conditions, they can also trigger unintended chemical pathways that lead to the formation of molecular hydrogen (H₂). This chemically induced hydrogen can distort mud gas readings and obscure signals attributed to natural hydrogen sources, especially in H₂S-prone formations.

Triazine-based H₂S scavengers react with H₂S primarily through nucleophilic substitution, forming thiadiazine intermediates. However, at elevated temperatures (typically >120–150 °C), decomposition of the triazine ring can initiate redox reactions, particularly in the presence of catalytic surfaces such as iron or steel, which are abundant in downhole assemblies. These reactions can yield elemental sulfur (often visible as yellow precipitates) and molecular hydrogen as byproducts.

The extent of hydrogen generation depends on several factors:

- Thermal environment: HPHT conditions accelerate both triazine decomposition and redox activity.
- Scavenger dosage: Excess triazine may promote incomplete or side reactions, favoring H₂ production.
- Catalytic influence: Iron-bearing solids and tubulars can facilitate electron transfer steps.
- Fluid composition: Alkalinity and the presence of organic additives like glycols or amines can influence the stability of reactive intermediates.

Recent work by [Qubaisi et al. \(2024\)](#) demonstrated that triazine-based hydrogen sulfide (H₂S) scavengers, particularly in the presence of iron-bearing substrates, can generate detectable amounts of hydrogen gas (H₂) as a secondary reaction product. This represents a non-geologic source of hydrogen that must be accounted for when interpreting gas anomalies in mud gas logs, especially in sour reservoir environments. Neglecting scavenger-related hydrogen contributions may produce false positives or inflate estimates of native hydrogen content.

Thermal Degradation of Organic Drilling Additives

High-temperature breakdown of drilling fluid components, such as esters, lubricants, and emulsifiers, represents a third source of artificial hydrogen. OBM systems are particularly susceptible above 300°C, where pyrolysis of glycol-based and ester-based additives has been shown to produce hydrogen, methane, and alkenes in laboratory HPHT experiments (Okoro et al., 2016). While often secondary to DBM, this mechanism can still contribute significantly to background hydrogen levels in hot wells or under prolonged bottomhole exposure. Even if a water-based mud (WBM) system is used with significant amounts of temperature-degradable organic additives, the risk of artificial hydrogen generation increases.

Quantifying and distinguishing artificial hydrogen sources is essential for an accurate interpretation of mud gas logs in hydrogen exploration wells. As summarized in Table 2, these sources include drill-bit metamorphism, chemical reactions involving H₂S scavengers, and thermal degradation of mud additives, each with distinct activation conditions and diagnostic gas signatures. Failure to account for these effects may result in overestimation of in-situ hydrogen potential. Future work should focus on the development of correction models based on gas ratios, compound-specific markers, and integration with bit temperature and lithology data.

Table 2—Summary of key mechanisms responsible for artificial hydrogen generation during drilling operations. Each process is characterized by its activation conditions, typical gas-phase products, diagnostic indicators useful for field identification, and relevant literature references.

Mechanism	Activation Conditions	Major Gas Products	Diagnostic Indicators	References
Drill-Bit Metamorphism (DBM)	High bit RPM, OBM, frictional heating (250–1200°C)	H ₂ , ethene, propene, CO	Light olefins, CO in OBM	Ibrahim et al. (2022); Herbinet et al. (2007)
H ₂ S Scavenger Reactions	Reaction with triazine or aldehyde scavengers	H ₂ , sulfur species	Yellow elemental sulfur precipitates; scavenger inventory; H ₂ S-rich formations	El-Sayed et al. (2021); Al-Yami et al. (2019)
Thermal Additive Decomposition	HPHT (>300°C), OBM or complex mud systems	H ₂ , CH ₄ , alkenes	Additive formulations, pyrolysis byproducts	Okoro et al. (2016)

Conclusion

The successful exploration of natural hydrogen depends on our ability to detect, quantify, and distinguish formation-derived hydrogen from gases generated during drilling operations. This study provides a comprehensive evaluation of hydrogen behavior in drilling environments, emphasizing the interplay between degassing dynamics, analytical technologies, and artificial generation mechanisms.

Field and laboratory investigations demonstrate that hydrogen, despite its low solubility in both water-based and oil-based muds, exhibits a delayed degassing profile similar to that of methane and other light gases. This behavior supports the continued use of conventional mud gas extraction systems, provided they are calibrated with hydrogen-specific correction models based on empirical data.

Among the technologies assessed, Gas Chromatography–Mass Spectrometry (GC-MS) emerges as a particularly robust and versatile solution. GC-MS enables the precise quantification of hydrogen across a broad concentration range, while simultaneously detecting key hydrocarbons and providing a qualitative assessment of the byproducts of drill-bit metamorphism. Its ability to operate without specialized carrier gases and maintain stability under field conditions makes it well-suited for integrated mud gas analysis during hydrogen exploration.

This work also highlights the need to account for artificial hydrogen generated through processes such as drill-bit metamorphism and thermal degradation of drilling additives. The co-occurrence of hydrogen with specific gas-phase markers enables identification and correction of non-formation contributions.

Controlled rig experiments reinforce this diagnostic approach, demonstrating consistent relationships between mechanical drilling conditions and the generation of artificial gases.

In summary, this study establishes a practical framework for incorporating hydrogen detection into existing mud logging operations. Through calibrated degassing workflows, multipurpose analyzers such as GC-MS, and diagnostic correction strategies, reliable hydrogen logging can be achieved using current infrastructure. As natural hydrogen exploration advances, these methodologies offer a scalable foundation for accurately assessing this promising clean energy resource.

References

- Larin, N., Zgonnik, V., Rodina, S., Deville, E., Prinzhofer, A., & Larin, V. N. (2015). Natural molecular hydrogen seepage associated with surficial, rounded depressions on the European craton in Russia. *Natural Resources Research*, **24**, 369-383.
- Prinzhofer, A., Moretti, I., Françolin, J., Pacheco, C., d'Agostino, A., Werly, J., & Rupin, F. (2019). Natural hydrogen continuous emission from sedimentary basins: The example of a Brazilian H₂-emitting structure. *International Journal of Hydrogen Energy*, **44**(12), 5676-5685.
- Zgonnik, V. (2020). The occurrence and geoscience of natural hydrogen: A comprehensive review. *Earth-Science Reviews*, **203**, 103140.
- Lollar, B. S., Onstott, T. C., Lacrampe-Couloume, G., & Ballentine, C. J. (2014). The contribution of the Precambrian continental lithosphere to global H₂ production. *Nature*, **516**(7531), 379-382.
- Lopez-Lazaro, C., Bachaud, P., Moretti, I., & Ferrando, N. (2019). Predicting the phase behavior of hydrogen in NaCl brines by molecular simulation for geological applications. *Bulletin de la Société géologique de France*, **190**(1).
- Wenger, Lloyd M., Robert J. Pottorf, Gordon Macleod, Glenn Otten, Sebastien Dreyfus, Holger Justwan, and Emily Sekula Wood. "Drill-bit metamorphism: Recognition and impact on show evaluation." In SPE Annual Technical Conference and Exhibition, SPE-125218. SPE, 2009.
- Herbinet, O., Marquaire, P. M., Battin-Leclerc, F., & Fournet, R. (2007). Thermal decomposition of n-dodecane: Experiments and kinetic modeling. *Journal of analytical and applied pyrolysis*, **78**(2), 419-429.
- Strapoc, D., Ammar, M., Abolins, N., & Gligorijevic, A. (2022, June). Key role of regearing mud gas logging for natural H₂ exploration. In SPWLA Annual Logging Symposium (p. D031S001R003). SPWLA.
- Qubaisi, K., Meridji, Y., Rubaie, N., & AlAkeely, A. (2024, February). Innovative Method to Predict H₂S Concentration from Advanced Mud Logs. In International Petroleum Technology Conference (p. D031S113R009). IPTC.